

Pinhole Precision Challenge

Build the sharpest camera image with no lens.

Win condition: Best image clarity and concept explanation after one aperture redesign.

THE SCIENCE YOU ARE USING

Light travels in straight lines. A pinhole camera has no lens. Light from each point of a scene travels in a straight line through a tiny hole and lands on the screen, forming an upside-down image. Because the rays cross at the hole, the picture is flipped.

The aperture tradeoff. A smaller hole makes a sharper image but a dimmer one; a bigger hole is brighter but blurrier. You optimize the aperture for the best balance of sharpness and brightness, then redesign once.

HOW TO BUILD AND RUN IT

- 1 Build the viewer.** Make a tube or box with a foil aperture at one end and a paper screen at the other.
- 2 Punch a small hole.** Pierce a clean, tiny hole in the foil with a pin. Smaller is sharper.
- 3 Find the image.** Aim at a bright scene and look for the dim, upside-down image on the screen.
- 4 Tune the aperture.** Try a slightly larger or smaller hole. Balance sharpness against brightness.
- 5 Explain the optics.** Describe why the image is flipped and why hole size changes the result.

Safety: No direct sun viewing. Pins handled at a controlled station. Allergy: None.

MATERIALS

Cardstock	shared
Foil	for aperture
Tape	shared
Pushpins or paper clips	shared
White screen paper	per team
Light source	shared

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why):

The ONE variable we are testing:

Baseline result:

After redesign:

DEFEND YOUR DESIGN

What evidence made you change your design?

If you ran it again, what is the ONE thing you would change?

HOW YOU SCORE (100 POINTS)

Scoring criterion	Points
Image clarity	35
Concept explanation	25
Build quality	20
Redesign	10
Speed	10
Total	100